
SU2QC Nine-Month Plan

Three nine-month ideas derived from the Genesis Phase-I proposal, the best first month for each, a rebuilt one-month project, and a complete plan for JEPA world models on SU(2) gauge dynamics · prepared 7 September 2026

Contents

1	Read this first	2
2	The four ideas at a glance	3
3	Idea 1: Matter on the plaquette	3
3.1	The open problem	4
3.2	What you build	4
3.3	Nine-month schedule	5
3.4	Gates and kill criteria	6
3.5	Best first month	6
4	Idea 2: Gauss-loop, symmetry-verified closed-loop execution	6
4.1	The open problem	6
4.2	What you build	7
4.3	Nine-month schedule	7
4.4	Gates and kill criteria	8
4.5	Best first month: Gauss-law syndromes on hardware	8
5	Idea 3: TN-Embed, hybrid quantum–classical embedding	9
5.1	The open problem	9
5.2	What you build	9
5.3	Nine-month schedule	9
5.4	Gates and kill criteria	10
5.5	Best first month: state-preparation distillation and the boundary map	10
6	Which first month to run first	11
7	The GPT-5 and Codex one-month project: judgment	11
7.1	What it gets right	11
7.2	Why it does not solve a physics problem	11
7.3	Verdict	11
8	The rebuilt one-month project: SU(2) with dynamical matter on one plaquette	12
8.1	Hypothesis	12
8.2	Model and observables	12
8.3	Claim boundaries	13
8.4	Thirty-day plan	13
8.5	Plan B	14
9	Gauge-JEPA: a complete plan for QML world models with JEPA	14
9.1	Why JEPA is the right world-model paradigm for quantum data	14
9.2	Definitions	15
9.3	Architecture	15
9.4	Training objective	15
9.5	Data	16
9.6	What the world model is for	16
9.7	Baselines and the honest advantage test	16
9.8	Nine-month arc	17
9.9	First month: Gauge-JEPA-0	17
9.10	Risks and mitigations	18
10	How the pieces fit together	18
11	Sources	19

1 Read this first

This plan assumes a nine-month window starting in October 2026 (month 1 = October), one lead researcher plus an advisor, IBM Quantum access (Heron-class devices and Nighthawk r2), one GPU node with CUDA-Q and cuTensorNet, and the software stack the GPT-5 monograph already established (Qiskit 2.x, PyZX, exact statevector references). Where a choice depends on something not stated in the four source documents, the choice is stated so it can be changed.

The four documents disagree about what the first project should be, and the disagreement is instructive. The proposal promises real-time confinement, string breaking, and meson and baryon production for $SU(2)$ in 2+1 dimensions on hardware within nine months. The monograph (GPT-5, 31 August 2026) picks a five-qubit pure-gauge plaquette chain and a compiler experiment, and says openly that string tension, string breaking, and hadronization are not supported by it. The roadmap (30 August 2026) recommends a physics-aware ZX compiler as the flagship and defers scientific QML to month 9. The repository report is a validation-hierarchy manual. The right reading is that the monograph and roadmap built an excellent verification scaffold around the wrong scientific object. This plan keeps the scaffold and changes the object.

One fact changes the target and must be taken into account. The proposal states that no non-Abelian $SU(2)$ gauge theory in 2+1 dimensions has been realized on quantum hardware. As of the past few months that is no longer true for the pure-gauge theory: plaquette chains of 5 to 151 qubits have been run on five IBM devices with entanglement-spectrum measurements (Chen et al., March 2026); adjoint string breaking of $SU(2)$ on 8×8 and 16×8 triangular-lattice plaquettes has been run on `ibm_boston` with circuits of up to 7,634 CZ gates at two-qubit depth 218, in quantitative agreement with matrix-product-state simulation (Ciavarella, de Putter, Younis, Rrapaj, 28 August 2026); and non-Abelian string breaking through gluonic screening has been observed on a three-plaquette ladder with trapped-ion qudits (John et al., May 2026). In 1+1 dimensions, $SU(2)$ with dynamical fermions has been run at 60 sites and 120 qubits in the loop-string-hadron basis with 17,660 two-qubit gates (Ilčić et al., February 2026). What, to our knowledge, has still never been run on a quantum processor is the thing the proposal actually promises: $SU(2)$ in 2+1 dimensions with dynamical fundamental matter, where a string breaks into mesons and baryon–antibaryon pairs and finite baryon density delays the breaking. That physics exists only in tree-tensor-network simulation (Cataldi, Orlando, Halimeh, September 2025) and in classical emulation of a small loop-string-hadron qubit model (Yang, Matsuda, Huang, Kashiwa, PRD August 2025). The nine-month program should be aimed squarely at that gap; everything else in this plan is built to make that shot succeed.

Target	Status, September 2026	Evidence
2+1D $SU(2)$ pure gauge, plaquette chains, real time	Done, up to 151 qubits on IBM	Chen et al. 2026
2+1D $SU(2)$ pure gauge on a 2D lattice, adjoint string breaking	Done, 16×8 plaquettes, 7,634 CZ	Ciavarella et al. 2026
Non-Abelian string breaking by gluons, ladder, qudits	Done, 3 plaquettes	John et al. 2026
1+1D $SU(2)$ with dynamical fermions, hadron dynamics	Done, 120 qubits	Ilčić et al. 2026

Target	Status, September 2026	Evidence
2+1D SU(2) with dynamical fermions, string breaking	Classical only (TTN 8×8 ; LSH emulation)	Cataldi et al. 2025; Yang et al. 2025
2+1D SU(2) with dynamical fermions on hardware	Open, to our knowledge; literature check at preregistration	—
Digitization with a demonstrated continuum limit (fuzzy gauge theory)	Open; conjectured, evidence incomplete by the authors' own account	Alexandru et al. 2024
Drift-adaptive, symmetry-verified closed-loop execution	Open; only static protections exist	Yao 2025; Sukeno et al. 2026
Hybrid tensor-network/QPU embedding for non-Abelian real time	Open	—
Learned world models on quantum-simulation data	Open; JEPA exists only for classical physics data	Phys-JEPA, JEPA-x 2026

A second correction matters for the encoding work. The proposal says the team's non-commutative-geometry truncation "is shown to have a correct continuum limit." That is demonstrated for the truncated sigma model, but for the fuzzy gauge theory itself the authors state that the theory is conjectured to lie in the same universality class as SU(2) and that a nonperturbative calculation providing clear evidence for the correct continuum limit has not yet been possible (Alexandru et al., PRD 109, 094502). That is not a weakness to hide; it is an open problem the team is uniquely placed to attack, and it is folded into Idea 1 below.

2 The four ideas at a glance

Idea	Object	Open problem	Deliverable at month 9	First month
1, Matter on the plaquette	Hardcore-gluon SU(2) with staggered fermions on $2 \times L$ ladders	No hardware run of 2+1D SU(2) with dynamical matter	Hardware string breaking with meson and baryon channels, verified against exact and TTN references	The rebuilt one-month project
2, Gauss-loop	Same circuits in syndrome-bearing and syndrome-free encodings	No drift-adaptive controller rewarded by gauge-law syndromes	A closed loop with a hardware-measured learning curve; whether gauge redundancy pays	Gauss-law syndromes on hardware
3, TN-Embed	Tensor-network teacher, QPU window on the string region	No hybrid partition with error accounting for non-Abelian real time	Distilled state preparation and a classical/quantum boundary map	Distillation and the boundary map
Gauge-JEPA	Latent world model with classical or quantum encoders	No world model for quantum-simulation data	Forecasting, latent planning, hardware twin; honest quantum-versus-classical test	Gauge-JEPA-0 on exact 1+1D data

3 Idea 1: Matter on the plaquette

3.1 The open problem

The proposal’s central physics deliverable, confinement and string breaking of $SU(2)$ in 2+1 dimensions on hardware, is now specifically the with-matter problem. Pure-gauge chains and lattices have been run at scale, and gluonic string breaking has been seen on qudits. To our knowledge, and subject to a literature check at preregistration, nobody has evolved $SU(2)$ in two spatial dimensions with dynamical fundamental fermions on a quantum processor. The classical benchmark is unusually clean: Cataldi, Orlando and Halimeh found that strings break at the resonance $m = 3g_E/8$, the mass at which, in their normalization $H_E = g_E \sum \mathbf{E}^2$, the electric energy released by removing one link of $j = 1/2$ flux ($3g_E/4$) matches the cost $2m$ of one quark–antiquark pair; in the proposal’s normalization $(g^2/2a) \sum \mathbf{E}^2$ the same resonance sits at $m = 3g^2/(16a)$. Breaking proceeds through both meson and baryon–antibaryon channels, which is a distinctively non-Abelian signature, and finite baryon density delays breaking (baryon blockade). Those are the observables a hardware run must reproduce.

3.2 What you build

The model is the Kogut–Susskind Hamiltonian of the proposal’s Eq. (1) with two-color staggered fermions, in the hardcore-gluon truncation $j \in \{0, \frac{1}{2}\}$, on the smallest geometries that contain both plaquettes and matter: a 2×2 patch (one plaquette, four matter sites), then 2×3 (two plaquettes), 2×4 , and 2×5 or 2×6 ladders. A spin-network count of gauge-invariant states, done for this plan, gives 82 states for the 2×2 patch, 1,727 for 2×3 , and 37,165 for 2×4 at $j_{\max} = 1/2$, and 152 and 8,225 for 2×2 and 2×3 at $j_{\max} = 1$. The count grows by roughly a factor of 21 per added column, so exact (Krylov) references are available through 2×6 (about 2×10^7 states), which covers every system that can reach hardware within nine months; the tree-tensor-network method of Cataldi et al. is needed for 2×7 and beyond and for wider patches.

A vertex with two links (the four corners of any ladder) has 6 gauge-invariant local states and fits in 3 qubits; a vertex with three links (every interior ladder vertex) has 13 and needs 4 qubits. The local encoding therefore costs 12 qubits for 2×2 , 20 for 2×3 , 28 for 2×4 , 36 for 2×5 and 44 for 2×6 .

Encoding	Qubits for 2×2 / 2×3	What it gives	Role
Local gauge-invariant (loop-string-hadron on a general graph, or dressed-site basis)	12 / 20	6 or 13 physical states per vertex in 3 or 4 qubits; the unused states and the link-consistency constraints are free leakage flags	Primary hardware encoding
Gauge-redundant Kogut–Susskind (links as 5-state electric basis, matter by Jordan–Wigner)	20 / 33	Full non-Abelian Gauss-law syndromes at every vertex	Syndrome study for Idea 2; exact cross-check of the primary Hamiltonian
Compact global (82 states in 7 qubits)	7 / 11	Fewest qubits, dense non-local Hamiltonian	Simulator-only benchmark

Encoding	Qubits for 2×2 / 2×3	What it gives	Role
Fuzzy gauge theory (16-dimensional link space, 4 qubits per link)	24 / —	A digitization with a conjectured continuum limit; the authors quote 236 CNOT and 88 rotations per plaquette	Classical truncation cross-check at 2×2 in months 2 and 8
q-deformed $SU(2)_k$ ($k = 2$ is what John et al. ran on qubits; baseline circuits by Webb-Mack and Klco)	depends on k	An alternative finite regularization with its own circuit primitives	Optional comparison at 2×2 if month 2 has slack

The primary observables are the Casimir profile along the string, the site-resolved matter density, projectors onto the intact string, the meson channel, and the baryon–antibaryon channel, string survival, breaking time as a function of mass around the resonance, and baryon blockade seeded by initial baryon density. Energy conservation and Gauss-law or leakage-flag violation are first-class outputs, following the repository report’s insistence that gauge violation be promoted from an implementation detail to a measured quantity.

The hardware path is 12 and 20 qubits on a Heron-class device, then 28 to 44 qubits on Nighthawk r2, whose square-lattice coupling map is a much better match for three- and four-qubit vertex blocks arranged as a ladder than heavy-hex is, although the routing overhead has to be measured in month 2 (IBM Nighthawk r2, 31 August 2026). The figure of merit throughout is two-qubit depth per qubit, not total gate count. The Ciavarella et al. run (7,634 CZ over 156 qubits at two-qubit depth 218) is the existence proof that the mitigation stack (dynamical decoupling, Pauli twirling, readout extinction, operator decoherence renormalization, zero-noise extrapolation, symmetry averaging, ancilla post-selection) can carry $SU(2)$ circuits of depth about 200 to quantitative agreement with tensor networks; the budgets below stay inside that depth.

3.3 Nine-month schedule

Month	Work	Exit evidence
1	The rebuilt one-month project: 2×2 patch Hamiltonian by two independent routes, exact dynamics, encodings, circuits, pilot and first full hardware run	See the rebuilt month below
2	2×3 and 2×4 exact dynamics; $j_{\max} = 1$ and fuzzy cross-checks at 2×2 ; compile pipeline frozen (Strang with gauge-invariant term grouping, PyZX topology-preserving pass, heavy-hex and square-lattice routing)	Truncation-error statement; CZ per Strang step and two-qubit depth table for 2×3 and 2×4
3	Exact Krylov references for 2×5 and 2×6 ; MPS along the ladder and TTN for 2×7 to 2×12 , validated against exact 2×6 ; string-breaking resonance and channel split at ladder sizes; preregistration of the hardware program	TN agrees with exact to 10^{-3} at 2×6 ; preregistration signed

Month	Work	Exit evidence
4	Hardware production on the 2×2 patch with full statistics, two mitigation arms and a second calibration window: pair-creation onset, Casimir reduction, meson-versus-baryon split, leakage-flag post-selection	Gate 2: agreement with exact within 2σ over at least four time points
5	2×3 on Heron-class hardware; mitigation ablations; Idea 2's syndrome reward used to choose execution settings	First two-plaquette with-matter time series
6	2×4 or 2×5 on Nighthawk r2; baryon-blockade protocol (seed one baryon, measure breaking delay)	Blockade delay visible above error bars
7	Compiler learning-curve metric (proposal Section 7) on the frozen circuit family: RL or learned ZX rewriting versus Qiskit level 3 and PyZX, judged after routing and by observable error	Learning curve reported with equal-shot baselines
8	Largest feasible ladder; TN comparison; truncation-sensitivity statement using $j_{\max} = 1$ and fuzzy references	Error budget decomposed into truncation, Trotter, compile, device, shot
9	Paper: first quantum-hardware simulation of $SU(2)$ string breaking with dynamical matter on plaquette geometries in two spatial dimensions; code and data release	Manuscript with claim table and archive

3.4 Gates and kill criteria

Gate 1 at the end of month 2: one Strang step for 2×3 compiles to at most about 500 CZ on the target coupling map, with two-qubit depth per step at most about 60, and Trotter error below 5 percent on the primary observable across the resonance time window; otherwise the ladder program shrinks to the 2×2 patch plus a pure-gauge static-charge ladder and the with-matter claim is confined to one plaquette. Gate 2 at the end of month 4: the hardware 2×2 production run reproduces the exact matter density within 2σ over at least four time points with post-selection retaining at least 20 percent of shots; otherwise months 5 to 8 become a mitigation-and-encoding study rather than a scaling study, reported as such.

3.5 Best first month

The best first month is the rebuilt one-month project described below: it is the decisive de-risking step, because every later month depends on whether the 12-qubit patch compiles to a depth today's hardware can carry, and it produces the first with-matter data point on hardware.

4 Idea 2: Gauss-loop, symmetry-verified closed-loop execution

4.1 The open problem

The proposal's second objective is a closed loop between hardware and AI, with hardware calibration data collected on a two-hour cadence over thirty-day windows and a learning-curve metric as the decision gate. The physical failure that decides whether a lattice-gauge simulation is trustworthy is loss of gauge invariance. With gauge-invariant term grouping, in which every product-formula factor is itself gauge invariant (an electric term, a covariant hopping term, a

plaquette) and each grouped factor is synthesized as an exact block unitary rather than re-split into its non-commuting Pauli strings, Trotter error never leaves the physical subspace and every Gauss-law violation measured on the device is hardware-induced; a noiseless simulation of the compiled circuit must return zero violation and zero leakage, and that is a unit test of the compile pipeline. Coherent gauge drift from the product formula appears only when the splitting is done at the level of individual Pauli strings. That separation is what makes the syndromes a clean reward. The remedies in the literature are static: post-selection on Gauss’s law, symmetry protection, gauge-covariant codes, quantum error-correction codes tailored to truncated $SU(2)$ with vertex constraints as stabilizers (Yao, November 2025), and measurement-based simulation that reads one-form-symmetry syndromes at no extra cost, demonstrated for a $\mathbb{Z}_2$ theory on Quantinuum H2 (Sukeno, Rinaldi, Okuda, August 2026). Nobody has built a controller that uses on-device physics syndromes as its reward and adapts execution across calibration drift. The syndromes need no exact answer, so the loop keeps working at sizes beyond exact diagonalization, which is precisely where the proposal’s learning-curve metric must eventually be measured.

The monograph’s gauge-reduced chain cannot be the vehicle for this idea: in a basis where every computational state is physical there is nothing to detect. Idea 2 therefore requires the syndrome-bearing encodings from Idea 1 and turns the encoding comparison into an experiment of its own: is gauge redundancy worth its qubits on today’s hardware?

4.2 What you build

Two encodings of the same dynamics run side by side: a syndrome-free reference (the one-qubit-per-plaquette chain, and the fully gauge-fixed two-plaquette Hamiltonian that has been reported on IBM hardware by Froland, Grabowska and Li) and a syndrome-bearing encoding (the local gauge-invariant or gauge-redundant Kogut–Susskind bases of Idea 1). A syndrome dashboard measures Gauss-law violation ε_G , energy drift $\Delta E/E_0$, mirror or lattice-symmetry asymmetry A_{mirror} , and leakage-flag rate ℓ . The Trotter step and term ordering are fixed outside the loop from exact or tensor-network references, because the syndromes cannot see product-formula error and a controller allowed to choose the step would simply choose the coarsest one. The controller (contextual bandit or Bayesian optimizer first, reinforcement learning over compile choices later) selects the physical-qubit path, dynamical-decoupling sequence, twirling, zero-noise-extrapolation factors, PyZX pipeline, and whether to insert mid-circuit syndrome checks with reset, to maximize the reward

$$R = -(w_G \varepsilon_G + w_E |\Delta E|/E_0 + w_M A_{\text{mirror}} + w_L \ell) - w_S \cdot \text{shots},$$

with the weights fixed in advance by regression of syndromes against exact observable error at exactly diagonalizable sizes. The proposal’s thirty-day calibration campaign supplies the drift model: calibration features on a two-hour cadence predict syndrome rates, and the controller is tested on later days than it was trained on, following the roadmap’s Path G rule.

4.3 Nine-month schedule

Month	Work	Exit evidence
1	Gauss-law syndromes on hardware (first-month project below)	Syndrome–error correlation established at 2 and 3 plaquettes
2	Thirty-day calibration campaign starts; drift model from calibration features to syndrome rates	Held-out-day prediction of ε_G within its bootstrap interval
3	Bandit over execution settings with the syndrome reward, calibration window one; equal-shot comparison against a fixed tuned stack	Controller beats the fixed stack in window one
4	Dynamic circuits: mid-circuit syndrome checks with reset on Heron-class and Nighthawk r2 hardware; bandit repeated in calibration window two; post-selection yield versus depth	Gate 2: controller beats the fixed stack in both windows; yield-versus-fidelity curve
5 to 6	Reinforcement learning over compile choices (ZX rewrite family, routing, gate scheduling) rewarded after routing by syndromes, with the product formula itself held fixed; the proposal’s learning curve	Learning curve versus iteration with equal-shot baselines
7	Scale to systems whose physical dimension exceeds exact reach (of order 10^8 states and above): pure-gauge wider patches in the redundant encoding first, then Idea 1’s largest ladder as it becomes available in month 8; verification by invariants and tensor networks only	Controller output consistent with TN where TN converges
8	Transfer across devices (Heron, Nighthawk, optionally IonQ) and across the with-matter circuits of Idea 1	Transfer statement with distribution tails, not only means
9	Paper: symmetry-verified closed-loop execution of non-Abelian gauge dynamics	Manuscript and controller release

4.4 Gates and kill criteria

Gate 1 at the end of month 1: syndromes correlate with exact observable error at exactly diagonalizable sizes with rank correlation above 0.7; otherwise the reward is not usable and the program pivots to energy-drift-only control. Gate 2 at the end of month 4: the controller beats the fixed stack at equal total shots in both calibration windows (month 3 and month 4) without reversing sign in either; otherwise the result is published as a characterized null with the drift dataset.

4.5 Best first month: Gauss-law syndromes on hardware

Days	Work	Exit evidence
1 to 5	Build the syndrome-free and syndrome-bearing encodings for the two-plaquette model and a three-plaquette ladder; derive Gauss-law, leakage, energy, and mirror operators and their measurement groupings; exact checks	Operators verified against the projected exact Hamiltonian

Days	Work	Exit evidence
6 to 10	Noisy simulation with device noise models; correlation of each syndrome with exact observable error across depth, time, and noise strength	Correlation table; preliminary reward weights
11 to 15	Hardware runs in three calibration snapshots on one frozen qubit path, both encodings, same shots	Syndrome time series with job and calibration identifiers
16 to 20	Post-selection yield versus fidelity gain; redundancy cost accounting (qubits, depth, shots)	The answer, at 2 to 3 plaquettes, to whether redundancy pays
21 to 25	First bandit over path, decoupling, twirling, and extrapolation factors using the syndrome reward, judged against exact error	Bandit regret versus the exact-error oracle
26 to 30	Report, dataset schema for the thirty-day campaign, preregistration of month 2	Research note and schema

5 Idea 3: TN-Embed, hybrid quantum–classical embedding

5.1 The open problem

The proposal’s third objective is a hybrid scheme in which the quantum device simulates a localized region while the surroundings are treated classically. In real-time string breaking, entanglement grows in the string region and defeats matrix-product and tree-tensor-network methods at late times, while today’s hardware cannot carry global deep circuits. What is missing for a non-Abelian theory is a partition with error accounting: a tensor-network teacher that prepares the confined initial state and hands it to the processor as a shallow circuit, a quantum window covering only the string patch, a boundary treatment for the rest, and a map that says where classical reach ends. The roadmap’s Path F and Section 10 describe the ingredients; this idea assembles them into one experiment.

5.2 What you build

A tensor-network layer (MPS along the ladder, TTN for wider patches, run through cuTensorNet and CUDA-Q) produces converged initial states and reference dynamics, with the bond-dimension sweep and convergence table the roadmap prescribes. A distillation layer compiles the initial confined state into a shallow circuit by matrix-product-state-to-isometric-circuit synthesis and symmetry-preserving adaptive ansätze, with fidelity, observable error, and CZ count traded off explicitly. An embedding layer runs Trotter evolution on the quantum window only, with three boundary treatments compared: frozen, mean-field, and tensor-network-updated. The product is a scaling law: observable error as a function of window width and depth, with the classical boundary map $\chi_\varepsilon(N, t, W)$ stated for every published plot.

5.3 Nine-month schedule

Month	Work	Exit evidence
1	State-preparation distillation and the boundary map (first-month project below)	Circuits and map delivered
2	Distillation on $2 \times L$ ladders with matter; symmetry-preserving layers; transfer of circuits across couplings	Fidelity at least 0.99 at half the CZ count of a generic ansatz
3	Window embedding on simulators: error versus window width for the three boundary treatments	Boundary-treatment decision with error accounting
4	Hardware state preparation and short evolution for 1+1D loop-string-hadron chains of 8 to 16 sites	Prepared-state observables within 2σ of TN
5	Ladder window on hardware, using Idea 1's compile pipeline and Idea 2's execution settings	First hybrid non-Abelian window run
6	CUDA-Q automation of the teacher-student loop (the proposal's NVIDIA alignment)	Reproducible pipeline from tolerance to circuit
7	Scaling law: quantum-window size versus observable error at fixed depth	Law with uncertainty across sizes and couplings
8	Combination with Idea 1's largest ladder; where the window beats global evolution at equal depth	Pareto plot of error versus depth
9	Paper: tensor-network-embedded quantum windows for non-Abelian string dynamics	Manuscript and pipeline release

5.4 Gates and kill criteria

Gate 1 at the end of month 2: distilled circuits reach fidelity at least 0.99 with at most half the CZ count of a hardware-efficient ansatz at equal fidelity; otherwise the distillation is dropped and the program becomes a boundary-map and tensor-network validation study. Gate 2 at the end of month 5: window-embedding error is below Trotter error at equal depth; otherwise report where the hybrid fails and why.

5.5 Best first month: state-preparation distillation and the boundary map

Days	Work	Exit evidence
1 to 7	MPS and TTN of string states in 1+1D loop-string-hadron chains (16 to 64 sites) and $2 \times L$ ladders (L at most 8); Schmidt spectra, entanglement growth, the $\chi_\epsilon(N, t, W)$ map with runtime and memory	Convergence table for every published number
8 to 18	Distillation of confined initial states into circuits: isometric MPS compile versus symmetry-preserving adaptive ansatz; fidelity, observable error, CZ count curves; exact verification	Trade-off curves; circuits archived
19 to 25	Window embedding on simulators: window width versus error at fixed depth for frozen, mean-field, and TN-updated boundaries	Boundary comparison table
26 to 30	Report and CUDA-Q automation of the sweep	Research note and scripts

6 Which first month to run first

If one person runs the program, the order is Idea 1's month, then Idea 2's, then Idea 3's. Idea 1's month is the critical path for the proposal's headline deliverable and settles the encoding-and-depth question everything else depends on. Its pilot run also produces the first leakage-flag data on hardware, which Idea 2's month reuses; the non-Abelian Gauss-law syndromes of the redundant encoding need their own hardware time, so the two months share circuits and infrastructure but not all of their device hours. Idea 3's distillation becomes necessary only when the ladder reaches 2×4 and beyond, which is month 5 or 6 of Idea 1. If two or three people are available, the three first months run in parallel, because Idea 2's month needs mostly hardware time and Idea 3's month needs only GPU time.

7 The GPT-5 and Codex one-month project: judgment

7.1 What it gets right

The monograph is a model of how to run a hardware experiment: it separates Trotter, compiler, and device error and verifies the compiler layer exactly; it preregisters five gates and a claim table that lists what the experiment cannot establish; it freezes the qubit path, seeds, shots, and mitigation so that compiler variants are compared fairly; it reconstructs the full energy from six measurement bases to 10^{-10} ; it handles readout-mitigated quasiprobabilities correctly; it contains a genuine compiler finding, namely that the topology-preserving PyZX pass cut 118 native ECR gates to 94 while unrestricted full reduction inflated the routed count to 180; and its code base (Hamiltonian builder, Strang circuits, PyZX pipelines, dry-run-first hardware submission, bootstrap analysis, tensor-network cross-checks) is directly reusable. Every part of that scaffold survives into the rebuilt month.

7.2 Why it does not solve a physics problem

The scientific object is a five-plaquette pure-gauge chain at $j_{\max} = 1/2$: 32 states, exactly solvable, no matter, no string, no continuum control. The monograph's own claim table says string tension, string breaking, and hadronization are not supported. The same Hamiltonian family has already been run at 5 to 151 qubits on five IBM devices with entanglement-spectrum and anti-flatness measurements (Chen et al., March 2026), so a five-qubit run in October 2026 adds no physics. The AI component is a random forest choosing among four fixed, exact pipelines, with an expected effect of roughly 15 percent in native two-qubit count; it exercises neither the proposal's closed loop nor its learning-curve metric. The gauge-reduced basis has no Gauss-law syndrome, so the experiment cannot teach anything about gauge-error physics, which is the question that decides feasibility for the with-matter problem. And the hypothesis is about the compiler: a null result is honestly reportable, as the monograph says, but it does not move the program's physics deliverable. The roadmap's Path A, learned ZX rewriting evaluated after routing, is where the compiler novelty actually lives; the selector is a placeholder for it.

7.3 Verdict

Keep the scaffold, change the object, and demote the compiler comparison to a secondary endpoint that costs nothing once the circuits exist.

8 The rebuilt one-month project: SU(2) with dynamical matter on one plaquette

8.1 Hypothesis

For the hardcore-gluon SU(2) single-plaquette patch with four staggered two-color fermion sites (82 gauge-invariant states, 12 qubits in a local gauge-invariant encoding), a symmetry-verified second-order product-formula evolution on a Heron-class IBM device reproduces the exact pair-creation and Casimir-reduction dynamics of a flux string within 2σ over at least four time points, with leakage-flag post-selection retaining at least 20 percent of shots, at most about 250 CZ per Strang step, two or three Strang steps per circuit, at most about 1,000 CZ per circuit including state preparation and basis changes, and two-qubit depth below about 200. This would be the first quantum-hardware time series for a non-Abelian gauge theory with dynamical fermions on a plaquette geometry, the smallest two-dimensional unit; the phrase “2+1D” is reserved for the 2×3 ladder and beyond, where an interior vertex exists.

The problem this month answers is a current one: what does it cost, in qubits and two-qubit depth, to evolve SU(2) with fundamental matter on a plaquette on today’s hardware, and does the hardware reproduce the non-Abelian channel structure of string breaking with a decomposed error budget? Nothing larger is claimed. At this size the physics is exactly diagonalizable; the hardware does not add knowledge about SU(2), it establishes the resource frontier and the verification method that the proposal’s months 4 to 8 need. That is the honest version of what a one-month hardware project can solve.

8.2 Model and observables

The Hamiltonian is the proposal’s Eq. (1) on one square plaquette with open boundaries: electric term $\frac{g^2}{2a} \sum_{\ell} \mathbf{E}_{\ell}^2$ with $\mathbf{E}^2 = j(j+1)$ on each of the four links, staggered mass $m \sum_{\mathbf{n}} (-1)^{n_x+n_y} \hat{\psi}_{\mathbf{n}}^{\dagger} \hat{\psi}_{\mathbf{n}}$, gauge-covariant hopping $\frac{1}{2a} \sum (\eta \hat{\psi}_{\mathbf{n}}^{\dagger} \hat{U}_{\ell} \hat{\psi}_{\mathbf{n}'} + \text{h.c.})$ across the four links, and the magnetic term $-\frac{1}{2g^2a} \text{Tr}(\hat{U}_{\square} + \hat{U}_{\square}^{\dagger})$. Each matter site holds two colors, so its four Fock states are the color-singlet vacuum, the doublet, and the doubly occupied color singlet, which in SU(2) is the baryon. The truncation is $j \in \{0, \frac{1}{2}\}$. The gauge-invariant basis is built two independent ways, by spin-network coupling coefficients and by projecting the 20-qubit redundant Kogut–Susskind Hamiltonian onto its Gauss-law kernel, and the two must agree to 10^{-12} . Three limits are unit tests: $g \rightarrow \infty$ (pure electric), $m \rightarrow \infty$ (matter frozen; the plaquette must reduce to the monograph’s one-plaquette Hamiltonian $\tilde{H}_1$), and the ring limit without the magnetic term (a four-site periodic loop-string-hadron chain).

Staggering fixes the geometry of the initial state: a quark is a particle on an even vertex and an antiquark a hole on an odd vertex, and adjacent vertices of the plaquette have opposite parity, so a quark–antiquark pair sits on adjacent vertices and can be joined by the one-link path or the three-link path. The initial state is the stretched string, the pair joined by $j = 1/2$ flux along the three-link path. It has two competing decay channels: coherent shortening to the one-link path through the magnetic term, and breaking by pair creation on the two intermediate vertices (an even and an odd one) into two mesons. The baryon channel, a doubly occupied even vertex paired with a doubly empty odd vertex, is defined by projectors

and its weight is measured, as in Cataldi et al. The observables are the Casimir per link, the site-resolved matter density, projectors onto the stretched string, the short string, the meson channel and the baryon–antibaryon channel, string survival, the energy (conserved by the exact evolution, not by the product formula), and the leakage-flag rate. The mass is scanned across the finite-size resonance near $m = 3g^2/(16a)$ in the proposal’s normalization ($m = 3g_E/8$ in Cataldi’s), which exact diagonalization locates on day 5; the $j_{\max} = 1$ basis (152 states) gives the truncation error at the same couplings.

8.3 Claim boundaries

Supported if the gates pass	Not supported by this experiment
A 12-qubit symmetry-verified evolution of SU(2) with dynamical matter on a plaquette is within reach of current hardware at a stated depth and yield	String tension, a continuum limit, or hadron phenomenology
The meson-versus-baryon channel split, the string shortening, and the Casimir reduction survive on hardware within the error budget	Any classically intractable calculation; the patch is exactly solvable
An error budget decomposed into truncation, Trotter, compile, device, and shot contributions	Baryon blockade and the full 2+1D claim, which need ladders (Idea 1, months 5 and 6)
The compiler comparison of the monograph, as a secondary endpoint on physically meaningful circuits	Quantum advantage of any kind

8.4 Thirty-day plan

Days	Work	Exit evidence
1 to 3	Freeze conventions (units, staggered phases, qubit order, basis); build the 82-state Hamiltonian by both routes; run the three limit tests	Agreement to 10^{-12} ; limits pass; preregistration draft
4 to 6	Exact dynamics: string survival, shortening, Casimir reduction, meson and baryon projectors versus mass; locate the finite-size resonance; $j_{\max} = 1$ comparison	Truncation-error statement; time window chosen where physics is visible
7 to 9	Encodings: 3-qubit-per-vertex local encoding with leakage flags versus the 7-qubit compact encoding; Pauli decomposition, gauge-invariant term grouping with each grouped term synthesized as an exact block unitary, Strang circuits with exact unitary checks (monograph code reused); Trotter error versus repetitions	Circuit family verified; error falls as r^{-2} ; noiseless leakage rate exactly zero
10 to 12	Compile to heavy-hex targets with Qiskit level 3 and the monograph’s PyZX pipelines, judged after routing; noisy simulation with a device noise model; Plan B decision if the per-step budget is missed	CZ and two-qubit-depth table; pipeline chosen by routed count, the monograph’s finding applied
13 to 14	Measurement design (groupings for all observables), readout mitigation plan, post-selection on leakage flags and energy; preregistration signed	Six-to-twenty-setting measurement plan; frozen configuration

Days	Work	Exit evidence
15 to 17	CUDA-Q statevector and tensor-network cross-checks; full noisy pipeline rehearsal; decide the hardware window (two or three Strang steps, at most about 1,000 CZ per circuit, two-qubit depth below about 200)	Rehearsal output reproduces exact observables within the noise model
18	Dry run: inspect every ISA circuit, resource summary, approval token	Approval summary archived
19 to 20	Pilot: three times, all groupings, 2,048 shots; stop/go review on equivalence, mapping, bit order, symmetries	Pilot passes or defects listed
21 to 22	Fix only prespecified defects; re-freeze	Change log
23	First full run: five times, two mitigation arms (none; decoupling plus twirling plus decoherence renormalization), optional extrapolation arm; the month-4 production run repeats this with full statistics and a second calibration window	Job, calibration, usage records; raw immutable export
24 to 26	Analysis: exact comparison, bootstrap over time blocks, post-selection yield, error-budget decomposition; second calibration window if quota permits	Primary and secondary tables with intervals
27	Ablations: mitigation arms, encodings, compiler pipelines (the secondary endpoint)	Ablation table
28 to 30	Research note with claim table, limitations, and the 2×3 continuation plan; clean-environment replication	Release bundle

8.5 Plan B

If by day 12 the 12-qubit patch does not compile to a faithful window (Trotter error below 5 percent at a depth the noise model can carry), the month switches to the monograph’s own next step, a pure-gauge plaquette chain with two static fundamental charges, and measures the flux-tube energy and Casimir profile between the charges for separations of one to four plaquettes. That is physics-flavored, exactly verifiable, and reuses everything built by day 12, and the with-matter patch moves to month 2 with a reduced encoding.

9 Gauge-JEPA: a complete plan for QML world models with JEPA

9.1 Why JEPA is the right world-model paradigm for quantum data

A world model predicts how a system’s state changes under actions, in a learned latent space, so that an agent can forecast, plan, and detect surprises. Joint-embedding predictive architectures learn that latent by predicting the embedding of a future or masked observation from the embedding of the present one, never reconstructing the observation itself; collapse is prevented either by an exponential-moving-average target encoder with stop-gradient (I-JEPA, V-JEPA) or by an explicit distribution regularizer such as SIGReg, which [LeJEPA](#) introduced precisely to remove the moving-average target and its heuristics. Action-conditioned versions plan by optimizing action sequences in latent space with the cross-entropy method (V-JEPA 2), and the 2026 physics variants ground the latent in known physical variables to stop encoder-predictor co-adaptation ([Phys-JEPA](#), [PSG-JEPA](#), [JEPA-x](#)) and make the prediction probabilistic with physics constraints as priors ([VJEPA](#) and [Bayesian JEPA](#)).

For quantum-simulation data this is not one option among several; it is the natural fit. The observation is a quantum state that cannot be read out cheaply: reconstruction costs exponential tomography. A generative world model is therefore the wrong paradigm, while a predictive one in latent space needs only what an encoder emits, and for a quantum encoder that is a handful of expectation values measured with shots. In simulation the target encoder acts on the exact future state; on a processor the target is either an error-mitigated hardware latent at the later time or a simulator latent paired with a hardware context, which is the sim-to-real alignment the digital twin below is built on. Gauss’s law and energy conservation are exactly the constraint priors Bayesian JEPA wants. And the latent world model is the seed of the proposal’s long-term goal, a domain-specific foundation model that maps lattice-QCD problems onto quantum algorithms: a representation of gauge dynamics that transfers across couplings, sizes, and devices is the first thing such a model needs.

9.2 Definitions

The world is the SU(2) lattice gauge theory of Ideas 1 to 3: first the 1+1D loop-string-hadron chain with dynamical fermions (exact references up to 16 sites, MPS beyond), then the $2 \times L$ ladders. An action a_t is a control the experimenter can apply: a Trotter step of duration Δt at couplings (g, m), a quench of m or g, the insertion or removal of a static charge, and, when hardware latents are included, an execution setting from Idea 2. An observation o_t is what an encoder is allowed to see: in simulation, either the state $|\psi_t\rangle$ or a finite-shot measurement record in several bases; on hardware, only measurement records. The latent s_t has dimension 16 to 64 for the classical encoder and dimension k, between 8 and 32, for the quantum encoder, whose latent is its vector of measured expectation values.

9.3 Architecture

Two encoders are trained and compared throughout. The classical encoder is a symmetry-aware transformer or graph network over per-site gauge-invariant features computed from measurement records: loop, string, and hadron occupations, local electric energy, plaquette expectation, string projectors; gauge equivariance is built in the way lattice-gauge-equivariant networks build it ([gauge-equivariant graph networks](#), April 2026). The quantum encoder is a gauge-respecting quantum convolutional network W_θ appended to the state-preparation circuit, with convolution blocks acting on adjacent vertex blocks in the loop-string-hadron basis and pooling that discards redundant degrees while preserving total gauge charge (the roadmap’s Path D block structure, built with the equivariant-quantum-network constructions of [Nguyen et al.](#)); its latent is the vector of k expectation values $s^Q = (\langle \psi_t | W_\theta^\dagger O_j W_\theta | \psi_t \rangle)$ for $j = 1$ to k, with k between 8 and 32 gauge-invariant observables. The default anti-collapse mechanism is SIGReg on the latent distribution, which needs no moving-average target and, for the quantum encoder, avoids running a second circuit family for the target branch; the exponential-moving-average target with stop-gradient is kept as an ablation. The predictor P_ϕ is a small transformer over $(s_t, a_t, \dots, a_{t+k-1})$, and grounding heads h_q map latents to known physical quantities during training only.

9.4 Training objective

The loss has five parts. The prediction loss $\sum_k \|P_\phi(s_t, a_{t:t+k-1}) - E_{\text{target}}(o_{t+k})\|^2$ over horizons $k = 1$ to K , with teacher forcing early and free rollouts late, and with the target branch

under stop-gradient in the moving-average ablation. The anti-collapse regularizer, SIGReg or VICReg variance and covariance terms, applied to the residual part of the latent. The physics grounding loss $\sum_q \|h_q(s_t) - Q_q(o_t)\|^2$, with Q ranging over energy, total charge, total Casimir, and string length on exact data, and additionally Gauss-law or leakage violation on noisy and hardware data, where it is no longer identically zero; grounding makes the latent identifiable and removes co-adaptation. The dynamics-consistency loss, which enforces the semigroup property $P(2\Delta t) \approx P(\Delta t) \circ P(\Delta t)$ on the predictor (the roadmap’s Path H test) and penalizes drift of the predicted energy along rollouts. And, when both encoders are present, the cross-prediction loss of JEPA-x: the predictor advanced from the quantum latent must land on the classical target latent and vice versa, with the classical branch present only in training so that deployment can use the quantum encoder alone. Shot noise enters explicitly in simulation: the context latent is computed from S shots and the target from the noiseless state or a much larger shot budget, so the model learns to predict the clean future from noisy present measurements; on hardware the target is a mitigated hardware latent or a simulator latent, as stated above.

9.5 Data

Exact diagonalization generates trajectories for 1+1D chains of 8 to 16 sites and ladders 2×2 to 2×5 across couplings, masses, initial string lengths, and baryon densities, with random action sequences; a GPU produces 10^4 to 10^5 trajectories per day at these sizes. MPS and TTN references extend chains to 64 sites and ladders to 2×12 with the roadmap’s convergence tables. Hardware latents come from the runs of Ideas 1 and 2, tagged with calibration date. Splits follow the roadmap’s rule: held out by coupling, time window, lattice size, truncation, circuit seed, and device day, never by row.

9.6 What the world model is for

The first use is forecasting: from short-time latents, predict the string-breaking time and the meson and baryon yields at held-out couplings and sizes, compared with tensor-network truth, and measure how many shots the latent needs to reach a given forecasting error compared with direct estimation of the observables. The second is planning: choose ramps $g(t)$ and $m(t)$, quench protocols, or Trotter step schedules that reach a target latent, for example a confined string of given length at minimal energy or maximal screening at a given time, by cross-entropy-method optimization in latent space as V-JEPA 2 does, then verify the plan by exact diagonalization. This is quantum control through a learned world model, and it is exactly the state-preparation problem Idea 3 faces. The third is a digital twin: the residual between the latent predicted from ideal data and the latent measured on hardware is a physics-aware, label-free error signal that complements Idea 2’s syndromes and feeds the same controller; if hardware latents are included in training with calibration features as inputs, the twin predicts hardware outcomes for new circuits on new days.

9.7 Baselines and the honest advantage test

Every result is judged against the roadmap’s minimum suite: an autoregressive latent model without joint embedding, direct regression of observables from the same measurement records, ridge regression on the gauge-invariant features, the symmetry-aware graph network alone, classical shadows with the same number of shots, a classical simulation of the quantum

encoder circuit itself (shallow quantum convolutional networks that avoid barren plateaus tend to be classically simulable, and that baseline settles whether the quantum encoder adds anything), and tensor-network truth. The quantum encoder is allowed to claim anything only if, at a matched shot budget, its latent reaches a lower forecasting or probing error than the classical encoder on a task preregistered in month 4, with the number of tasks fixed in advance to keep the comparison honest; otherwise the paper reports the null and keeps the classical world model, which is still useful. End-to-end cost is reported as state preparation plus circuit executions plus shots plus optimization plus classical post-processing, following the roadmap’s Eq. (27).

9.8 Nine-month arc

Month	Work	Exit evidence
1	Gauge-JEPA-0: classical encoders on exact 1+1D data; anti-collapse, grounding, rollout and probing metrics; quantum-encoder stub	First-month project below
2	Scale to MPS data (chains to 64 sites); held-out coupling and size tests; probing of string-breaking time	Forecast within TN error at two held-out masses
3	Planning: ramp and quench design in latent space; verification by exact diagonalization; comparison with direct optimization	Planning success rate above greedy and random baselines
4	Quantum encoder on simulators: gauge-respecting QCNN trained with adjoint differentiation on statevectors (generalized shift rules or SPSA reserved for hardware, since the equivariant blocks have multi-level generators); shot-noise-aware loss; gradient-variance diagnostics; the advantage task preregistered	Trainable at 8 to 16 sites; gradient variance reported versus size
5	Cross-prediction between classical and quantum branches; the matched-shot advantage test including the classical simulation of the encoder circuit	Advantage table, positive or null
6	Hardware latents from Idea 1 and Idea 2 runs; twin residuals as an error signal; calibration-conditioned prediction	Residual detects a miscalibrated day with stated AUROC
7	Ladders: transfer from 1+1D to $2 \times L$, from small to large, with grounding on Casimir profiles	Transfer curve with intervals
8	Bayesian JEPA extension: probabilistic latents with Gauss’s law and energy as constraint priors; uncertainty calibration	Calibrated intervals on forecasts
9	Paper and release: a latent world model of non-Abelian gauge dynamics with quantum encoders	Manuscript, code, datasets

9.9 First month: Gauge-JEPA-0

Days	Work	Exit evidence
1 to 4	Exact 1+1D loop-string-hadron data: 12 to 16 sites, hardcore truncation, string initial states, actions Δt in $\{0.05, 0.1, 0.2\}$ and mass quenches; measurement records in three bases at 256 to 4,096 shots; splits by coupling and time window	Dataset with checksums and convergence of exact references
5 to 10	Classical Gauge-JEPA: feature extractor, transformer encoder, SI-GReg, predictor, grounding heads; moving-average target as ablation; training curves; collapse diagnostics (effective rank, isotropy)	Stable training; latent rank above 0.5 of dimension
11 to 15	Evaluation: rollout drift versus horizon, probing R^2 for string-breaking time and channel yields, held-out mass generalization; autoregressive and ridge baselines at equal shots	Preregistered metric table
16 to 20	Grounding ablation: with and without physics heads and semi-group loss; co-adaptation diagnostics	Ablation table
21 to 26	Quantum-encoder stub: gauge-respecting QCNN on an 8-site statevector, $k = 8$ observables, adjoint-differentiated, trained on the same loss; trainability only, no advantage claim	Loss decreases; gradient variance reported
27 to 30	Research note, code release, month-2 preregistration	Release bundle

9.10 Risks and mitigations

Collapse is caught by effective-rank and isotropy diagnostics and by the grounding heads; trivial latents that encode only energy are caught by the probing tasks; shot noise is handled by noisy-context, clean-target training in simulation and by mitigated targets on hardware; barren plateaus in the quantum encoder are mitigated, not removed, by shallow local convolutions, gauge-invariant pooling, and a curriculum from 8 sites upward, with gradient variance reported against size at every stage; co-adaptation is removed by grounding and cross-prediction; hardware drift is handled by conditioning on calibration features and holding out device days. The largest risk is scientific rather than technical: that the classical encoder, or a classical simulation of the quantum one, is simply as good. The plan is built so that outcome still leaves a useful world model and an honest paper.

10 How the pieces fit together

The four ideas share one infrastructure: the monograph’s verification scaffold, the Hamiltonian and encoding library that Idea 1’s first month produces, and the exact and tensor-network reference data that Ideas 1 and 3 generate and Gauge-JEPA consumes. Idea 3’s distilled state preparation lowers the depth Idea 1 must spend on hardware; Idea 2’s syndromes and Gauge-JEPA’s twin residuals feed the same controller; Gauge-JEPA’s planner designs the ramps Idea 3 prepares. The proposal’s decision-gate metrics are produced in specific places: the learning-curve metric by Idea 2 (with the compiler comparison as a secondary endpoint of Idea 1), the scaling metric by Idea 3’s boundary map, the physics metrics by Idea 1, and the foundation-model direction by Gauge-JEPA. A team of two can run Idea 1 and Gauge-JEPA as the main lines, with Idea 2’s and Idea 3’s first months as the second person’s months 2 and 3; a team of one should run Idea 1 to its month-4 hardware result before opening a second line.

11 Sources

Genesis Phase-I proposal, “AI-Accelerated Non-Abelian Gauge Dynamics on Quantum Hardware” (NMSU, BNL, UMD); the GPT-5 one-month monograph (31 August 2026); the SU(2) QML research roadmap (30 August 2026); the physics and mathematics report on two quantum-simulation repositories (31 August 2026).

- Chen, Chen, Meher, Müller, Schäfer, Yao, Thermalization of SU(2) lattice gauge fields on quantum computers, <https://arxiv.org/abs/2603.23948>
- Ciavarella, de Putter, Younis, Rrapaj, Quantum simulations of two-dimensional non-Abelian adjoint string breaking, <https://arxiv.org/abs/2608.28752>
- John et al., Non-Abelian string-breaking dynamics on a qudit quantum computer, <https://arxiv.org/abs/2605.05841>
- Ilčić et al., Observation of robust and coherent non-Abelian hadron dynamics on noisy quantum processors, <https://arxiv.org/abs/2602.18080>
- Cataldi, Orlando, Halimeh, Real-time string dynamics in a 2+1D non-Abelian lattice gauge theory, <https://arxiv.org/abs/2509.08868>
- Yang, Matsuda, Huang, Kashiwa, Quantum simulation of QC₂D on a two-dimensional small lattice, Phys. Rev. D 112, 034511 (2025), <https://journals.aps.org/prd/abstract/10.1103/pj6x-zymq>
- Rouxinol, Bender, Emonts, Grossi, Halimeh, Quantum phase diagram of the 2+1D untruncated SU(2) lattice gauge theory with dynamical fermions, <https://arxiv.org/abs/2607.26132>
- Froland, Grabowska, Li, Simulating fully gauge-fixed SU(2) Hamiltonian dynamics on digital quantum computers, <https://arxiv.org/abs/2512.22782>
- Gupta et al., String-breaking statics and dynamics in a (1+1)D SU(2) lattice gauge theory, <https://arxiv.org/abs/2603.24698>
- Gupta, Kadam, Bapat, Stryker, Davoudi, Raychowdhury, Tensor-network toolbox for probing dynamics of non-Abelian gauge theories, <https://arxiv.org/abs/2501.18301>
- Gauge loop-string-hadron formulation on general graphs, JHEP 12 (2025) 060, [https://link.springer.com/article/10.1007/JHEP12\(2025\)060](https://link.springer.com/article/10.1007/JHEP12(2025)060)
- Alexandru et al., Fuzzy gauge theory for quantum computers, <https://arxiv.org/abs/2308.05253>
- Yao, Quantum error correction codes for truncated SU(2) lattice gauge theories, <https://arxiv.org/abs/2511.13721>
- Sukeno, Rinaldi, Okuda, Measurement-based simulation of lattice gauge theory dynamics with adaptive quantum circuits on a trapped-ion processor, <https://arxiv.org/abs/2608.04290>
- Webb-Mack, Klco, Deforming the trail: baseline quantum circuitry for SU(2)_k lattice gauge theory, <https://arxiv.org/abs/2605.15076>
- IBM Quantum Nighthawk r2, <https://www.ibm.com/quantum/blog/nighthawk-r2>

- Gauge-equivariant graph neural networks for lattice gauge theories, <https://arxiv.org/abs/2604.20797>
- Nguyen et al., Theory for equivariant quantum neural networks, <https://arxiv.org/abs/2210.08566>
- Assran et al., V-JEPA 2, <https://arxiv.org/abs/2506.09985>
- LeJEPA and SIGReg, <https://arxiv.org/abs/2511.08544>
- Phys-JEPA, <https://arxiv.org/abs/2606.16076>
- PSG-JEPA, <https://arxiv.org/abs/2608.06799>
- JEPA-x, <https://arxiv.org/abs/2608.24044>
- Variational and Bayesian JEPA, <https://arxiv.org/abs/2601.14354>